AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 14 · STUDENT LESSON

# Build a Human Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

---

## Before you start

Week 13: agreement with examples does not prove a rule is complete.

Plan time to read, investigate and explain. Keep predictions separate from observations.

---

STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Construct a classifier with a stated input format, selected rule and recorded output. Try the recall before opening notes.

### Your learning target

I can construct a **classifier** with a stated **input** format, selected rule and recorded **output**.

I can trace valid, boundary and invalid **input**s through different paths.

I can distinguish an **input** check from **evidence** that a **prediction** is correct.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Classification maps features to a category using a **decision boundary**.



### Your target

LEARNING TARGET A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

Learner name:

**Readiness for the connected explanation**

A fitted rule predicts tall at a height of at least 6 cm and short otherwise. Predict at 5 and 6 cm. What should you check before applying it to a blank height?

**This week's end goal**

By the end, I can build a complete **classifier** that validates **input**s before predicting.

---

STUDENT LESSON · 2 OF 18

# Notice the evidence

> Investigate the supplied Build a Human Classifier case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

**Look Closely Before You Explain**

Pictures give clues; **evidence** tells us what the clues mean.



### Start with this puzzle

MYSTERY Mystery creatures can be sorted using visible features such as antenna count and spot pattern, with the decision rule drawn on a feature map.

**What do you notice?**

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

### Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

**A closer explanation**

Your task is to build a paper **classifier** that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers.

Use the threshold of 6 centimetres selected after adding the height 5 short example in Week 13. Name this selected model Tower1.

Write an **input contract**: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts.

It is not a claim that every accepted height appeared in **fitting** or that towers outside this range cannot exist.

A complete **classifier** needs instructions for checking the **input**, applying Tower1 when the **input** is accepted, and recording the outcome. It must also say what happens when the **input** fails a check.

The threshold alone does not provide these surrounding instructions.

---

STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Build a Human Classifier. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.



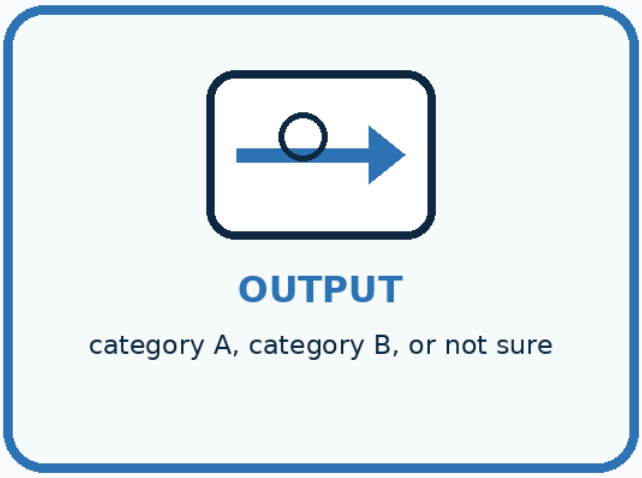
**INPUT**

creature features



**PROCESS**

compare the feature point with a boundary




**INPUT** creature features

**PROCESS** compare the feature point with a boundary

**OUTPUT** category A, category B, or not sure

### Key idea

CORE IDEA Classification maps features to a category using a **decision boundary**.

### How the mechanism works

UNDER THE HOOD A **classifier** partitions **feature space** into decision regions. A **decision boundary** is where the predicted category changes.

Different boundaries may fit the same **training** examples but behave differently on unseen or borderline **input**s.

Draw a fresh example with all three stages:

## A closer explanation

### Three records through the Tower1 classifier

| INPUT RECORD | CHECK AND RULE | RECORDED OUTCOME |
| --- | --- | --- |
| 5 cm | Accepted; 5 is below 6 | Tower1 predicts short |
| 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall |
| Height blank | Missing height; stop before rule | Tower1; no prediction |

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10.

If any check fails, record **no prediction** and the reason, then stop **process**ing that record. Do not continue into the threshold comparison.

If all checks pass, compare the height with 6. At least 6 gives the **prediction** tall; otherwise give short. Record the original **input**, **model version** Tower1 and the **prediction**.

For a rejected record, record the original **input**, Tower1, **no prediction** and a reason. Tower1 identifies the intended model even when its threshold was not used.

## Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

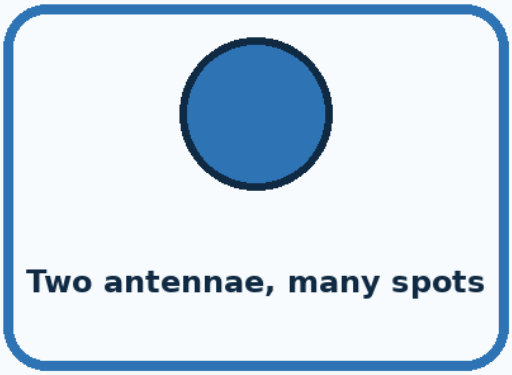
# Compare the cases

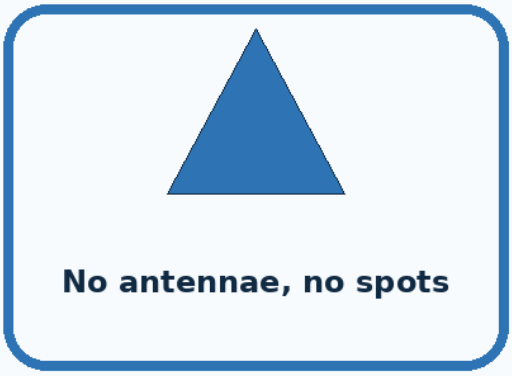
> Compare the supplied cases for Build a Human Classifier. Identify the detail that changes the conclusion.

PICTURE GALLERY

### Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









### The case to investigate

CASE 1 — Two antennae, many spots Clue: far inside region A. Meaning: stable A **prediction**.

### The case to investigate

CASE 2 — No antennae, no spots Clue: far inside region B. Meaning: stable B **prediction**.

### The case to investigate

CASE 3 — One antenna, some spots Clue: near boundary. Meaning: uncertain.

### The case to investigate

CASE 4 — New wing feature Clue: not represented on map. Meaning: boundary may be incomplete.

### Keep this boundary clear

BOUNDARY A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

Which two pictures are easiest to confuse? What **evidence** separates them?

**Change one thing in the pictured case**

### Change one thing

Change: Provide length 9 with unit mm.

### Predicted result

**Prediction**: Request clarification rather than predicting long.

### Why the result follows

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

STUDENT LESSON · 5 OF 18

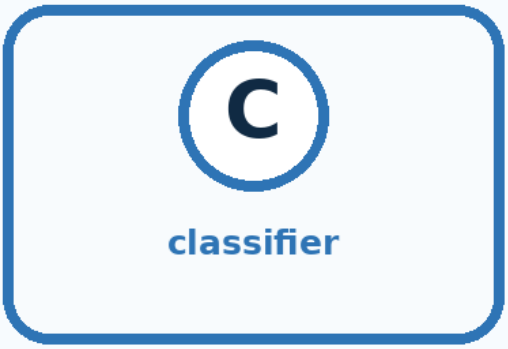
# Words that unlock the idea

> Use the vocabulary of Build a Human Classifier to explain a concrete decision, rather than only memorising definitions.

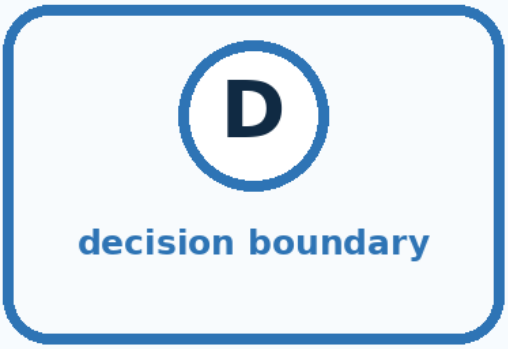
VOCABULARY

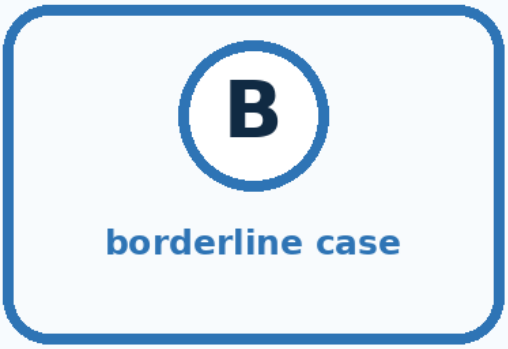
### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.






| Word | Meaning | My example |
|---|---|---|
| classifier | process that predicts a category | |
| feature space | map whose positions represent feature values | |
| decision boundary | place where predicted class changes | |
| borderline case | input near or across competing regions | |

### Misconception detective

#### Find the unsupported claim

CLAIM TO REPAIR Anything on the same side must truly belong to that category.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

### Words used in the closer explanation

| Word | Meaning |
| --- | --- |
| Input contract | The format and conditions an input must satisfy for a procedure to accept it |
| Validation | Checking whether an input satisfies the stated contract |
| No prediction | An explicit outcome when the classifier does not assign a category |
| Model version | A name that identifies the particular model used for a prediction |

---

STUDENT LESSON · 6 OF 18

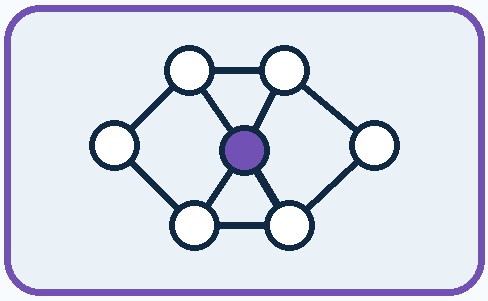
# Follow a worked example

> Follow the worked trace for Build a Human Classifier, predict a step, then compare with its explanation.

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



### The case to investigate

CASE Mystery creatures can be sorted using visible features such as antenna count and spot pattern, with the decision rule drawn on a feature map.

### Step 1

STEP 1 — OBSERVE creature features

### Step 2

STEP 2 — TRACE compare the feature point with a boundary

### Step 3

STEP 3 — CONCLUDE category A, category B, or not sure

### Model answer

MODEL REASONING Plot each creature by the selected features and draw a line or rule separating labeled examples. Cases near the line are sensitive to small changes and should receive careful review or a not-sure **output**.

### A claim to avoid

DO NOT OVERCLAIM A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

What new **evidence** could change this answer?

### A closer explanation

| Input | Contract result | Outcome |
|---|---|---|
| 1 cm | Accepted at lower endpoint | Predict short |
| 9 cm | Accepted | Predict tall |
| 10 cm | Accepted at upper endpoint | Predict tall |
| 0 cm | Outside 1 through 10 | No prediction |
| 60 mm | Unit is not cm | No prediction |
| 4.5 cm | Not a whole number | No prediction |

A valid 9 cm **input** receives a tall **prediction**, but the **input** check does not establish that the **prediction** matches its comparison label. **Validation** checks the contract.

Evaluation checks **prediction**s against appropriate **evidence**. Keep these questions separate.

### Worked explanation · Build a Human Classifier

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step.

Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Build a Human Classifier

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

### The facts we will use

#### Supplied facts

Valid **input**: whole-number length from 0 to 15 with unit cm.

#### Supplied facts

Predict long at least 9 cm; otherwise short. Missing, wrong-unit or out-of-range values request clarification.

AI ACADEMY / AI-1/14

# Build a Human Classifier

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Input contract

Check that length is present, whole, within 0-15, and in cm.



## 2. Valid cases

8 cm -> short; 9 cm -> long.



## 3. Invalid cases

16 cm and blank length -> clarification, not a forced label.



## 4. Keep records

Store input, contract result, rule version and prediction.

**CHANGE ONE THING**

Provide length 9 with unit mm.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Input contract | Check that length is present, whole, within 0–15, and in cm. |
| 2. Valid cases | 8 cm → short; 9 cm → long. |
| 3. Invalid cases | 16 cm and blank length → clarification, not a forced label. |
| 4. Keep records | Store input, contract result, rule version and prediction. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows


### Why the result follows


A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

### What this example does not establish

#### Keep the stated limit

Contract validity does not prove the predicted category agrees with a checked reference label.

#### Keep the stated limit

Read the labelled picture: Trace 0 cm and explain why clarification is different from predicting short.

#### Keep the stated limit

Change one thing: Provide length 9 with unit mm. Predict the result before checking the explanation. What remains the same?

---

STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Build a Human Classifier claim and explain a repair using the supplied evidence.

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

| Case | Evidence | What it shows |
|---|---|---|
| Two antennae, many spots | far inside region A | stable A prediction |
| No antennae, no spots | far inside region B | stable B prediction |
| One antenna, some spots | near boundary | uncertain |
| New wing feature | not represented on map | boundary may be incomplete |

### Repair two tempting claims

#### Challenge claim A

CLAIM A Anything on the same side must truly belong to that category.

Repair:

#### Challenge claim B

CLAIM B Only one possible boundary can fit the examples.

Repair:

#### A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

### A closer explanation

#### Check the claim against the facts

The **classifier** reports a **prediction** or **no prediction**. It does not decide what a person must do with a tower.

A separate classroom procedure might ask someone to remeasure a rejected **input** or compare a **prediction** with a recorded label. Write that action separately from the predicted category.

**No prediction** is not a third tower size. It records that this procedure did not assign short or tall.

Count and investigate such records when evaluating the whole system; hiding them could make a system look more useful than it is.

**A classifier accepts 9 cm and predicts tall. A learner says the input passed, so the label is definitely right. Correct the claim. Then explain why reporting short for a blank height would violate the Tower1 procedure**

---

STUDENT LESSON · 8 OF 18

## Practise together

Practise Build a Human Classifier with support, then explain the deciding step in your own words.

GUIDED PRACTICE

### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Classification maps features to a category using a **decision boundary**.

Use a fresh example to explain **classifier**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Only one possible boundary can fit the examples.

Invent one edge case and name the safe response.

### Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

#### A closer explanation

Write the procedure as numbered steps or connected boxes with labelled branches. Give it to a partner who follows only what you wrote.

The partner should record each check and stop point, including equality at the threshold and at both accepted range endpoints.

> **Construct Tower1 in your own words. Have a partner trace 1 cm, 6 cm, 10 cm, 11 cm and a blank height. Record each outcome and any instruction you had to clarify. Explain why the accepted 10 cm case still needs separate evidence of correctness**

A partner finding an unclear branch gives useful design feedback. Revise the instruction and retrace the affected examples.

Keep the model name and threshold visible so a change to the fitted rule is not confused with a correction to an instruction.

### Practise with less help: Construct both outcome paths

Use workbook W2 for the connected example “Constructing a **classifier**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Trace 0 cm and explain why clarification is different from predicting short.

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

---

STUDENT LESSON · 9 OF 18

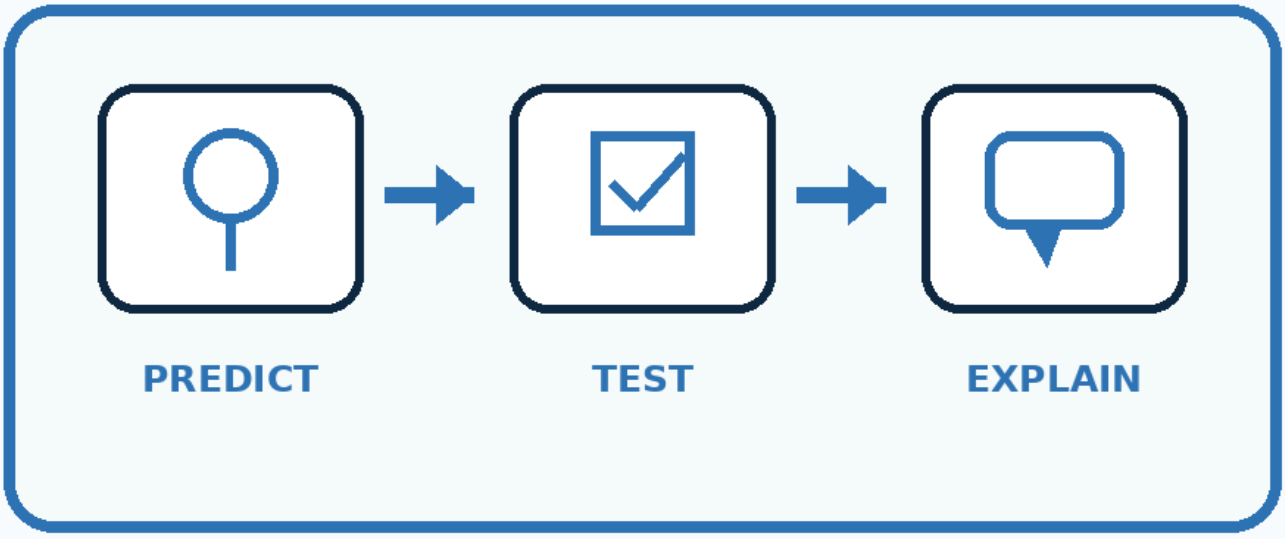
## Run the investigation

> Investigate Build a Human Classifier. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

### Mystery Creature Classifier

Predict first, test honestly, and explain what the **evidence** means.



### 1. Predict

State your **prediction** before the result is known.

### 2. Investigate

Complete the task: Classify mystery creatures and make the boundary visible.

### 3. Compare cases

Keep a normal case and an edge case clearly separated.

### 4. Record honestly

Record what happened—even when it does not match your **prediction**.

### 5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

| Case | Prediction | Observed result | What it means |
|---|---|---|---|
| Normal | | | |
| Edge | | | |

### Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

---

STUDENT LESSON · 10 OF 18

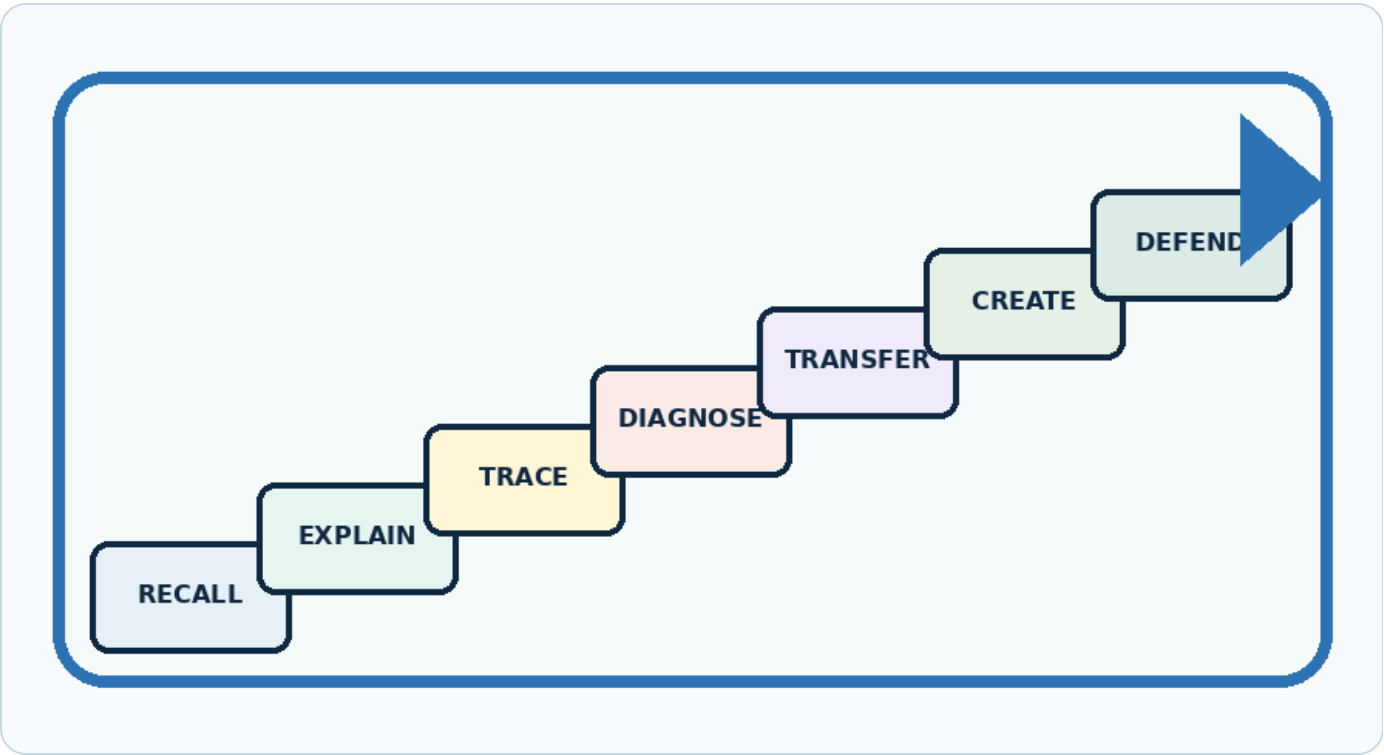
## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Build a Human Classifier. Explain why each step is justified.

PRACTICE LADDER

### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **classifier**.

EXPLAIN Classification maps features to a category using a **decision boundary**.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Anything on the same side must truly belong to that category.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

### When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

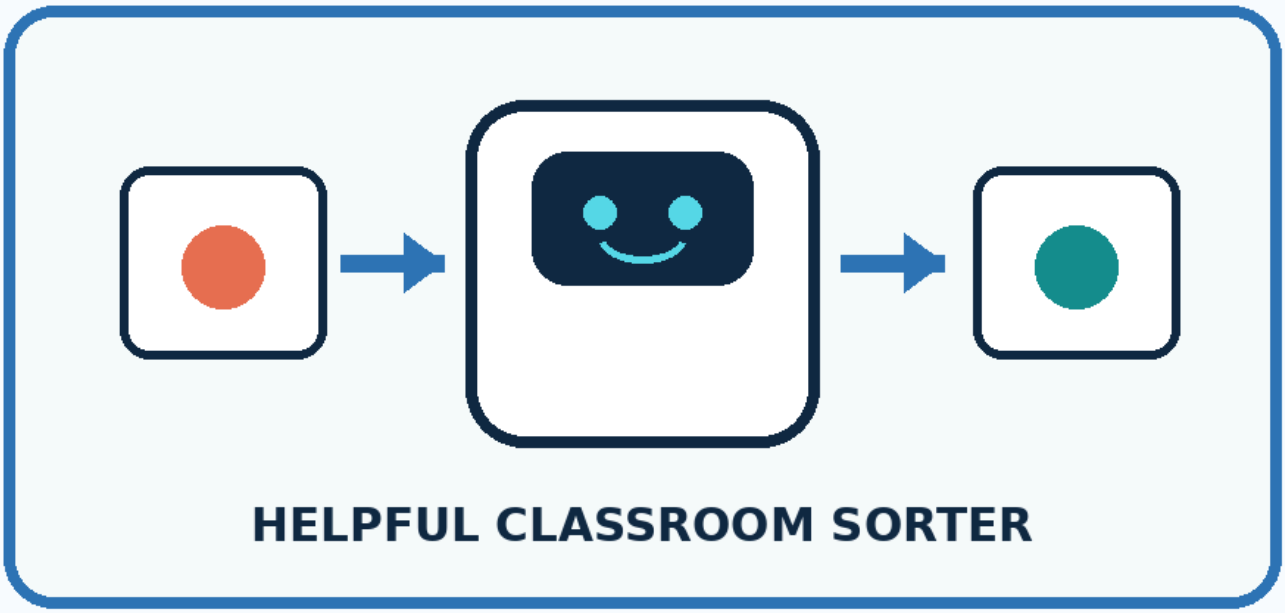
# Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.




> THIS WEEK'S CHECKPOINT Make the sorter's current **decision boundary** visible as a rule chart or feature map.

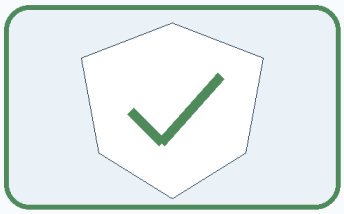
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



### Responsible design

RESPONSIBLE DESIGN A drawn boundary describes the **classifier**'s decision, not a natural law about the objects.

My project decision:

### Evidence for your project

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

---

STUDENT LESSON · 12 OF 18

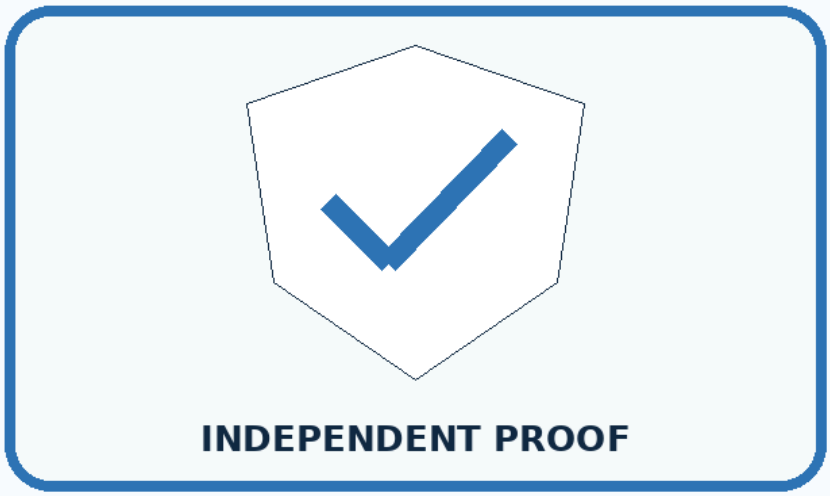
## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

### Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.




### What success looks like

REQUIRED PROOF Classify mystery creatures, draw the boundary, and explain two cases

whose category changes if the boundary moves.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

### Check your explanation

SELF-CHECK Plot each creature by the selected features and draw a line or rule separating labeled examples. Cases near the line are sensitive to small changes and should receive careful review or a not-sure **output**.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

### Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Save New Examples for Testing.

---

**STUDENT LESSON · 13 OF 18**

## Study a complete case

Revisit the complete Build a Human Classifier case for supported practice. Repeated examples do not become fresh assessments.

### Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

### The question

Apply the rule and identify a limitation.

### Worked explanation

The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

### Explain it in your own words

> Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### Trace the supplied case visually

### Does the object have wheels?

| Evidence or stage | Value or result |
| --- | --- |
| Bicycle | Yes → vehicle |
| Wheelchair | Yes → vehicle |

| Evidence or stage | Value or result |
| --- | --- |
| Toy car | Yes → vehicle |
| Boat | No → excluded by this rule |

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

Apply the rule and identify a limitation.

The first three receive vehicle; the boat does not. If the intended category includes boats, the rule misses them. Define the category independently of the **prediction** rule.

---

STUDENT LESSON · 14 OF 18

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Build an independent ribbon classifier

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Use a human **classifier**: three sides means triangle; four equal sides means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

### Connect to your project

Make the sorter's current **decision boundary** visible as a rule chart or feature map.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

### Transfer practice from the original programme

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

**Explain the mechanism, evidence and one limit.**

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

---

STUDENT LESSON · 15 OF 18

## Your lesson route

Use this map to navigate Build a Human Classifier, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional ribbon sorter uses a fixed instruction card called RibbonR.

| By the end, I can | How I will show it |
|---|---|
| Construct a classifier with a stated input format, selected rule and recorded output | Explain the deciding step, show a trace or test, and state one limit. |
| Trace valid, boundary and invalid inputs through different paths | Explain the deciding step, show a trace or test, and state one limit. |
| Distinguish an input check from evidence that a prediction is correct | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

---

**STUDENT LESSON · 16 OF 18**

## Finish and return

> Explain Build a Human Classifier without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Construct a **classifier** with a stated **input** format, selected rule and recorded **output**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

---

STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? Store **input**, contract result, rule version and **prediction**.

1. Invalid cases

2. **Input contract**

3. Keep records

4. Valid cases

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep records Store **input**, contract result, rule version and **prediction**. A **prediction** boundary answers a different question from **input validation**.

Length 9 passes the contract and meets the long boundary. Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Provide length 9 with unit mm. Which conclusion is supported by the supplied facts?

1. Request clarification rather than predicting long.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Request clarification rather than predicting long. RibbonR requires cm; the unit mismatch fails its declared **input contract**.

**Repair a missing building block**

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

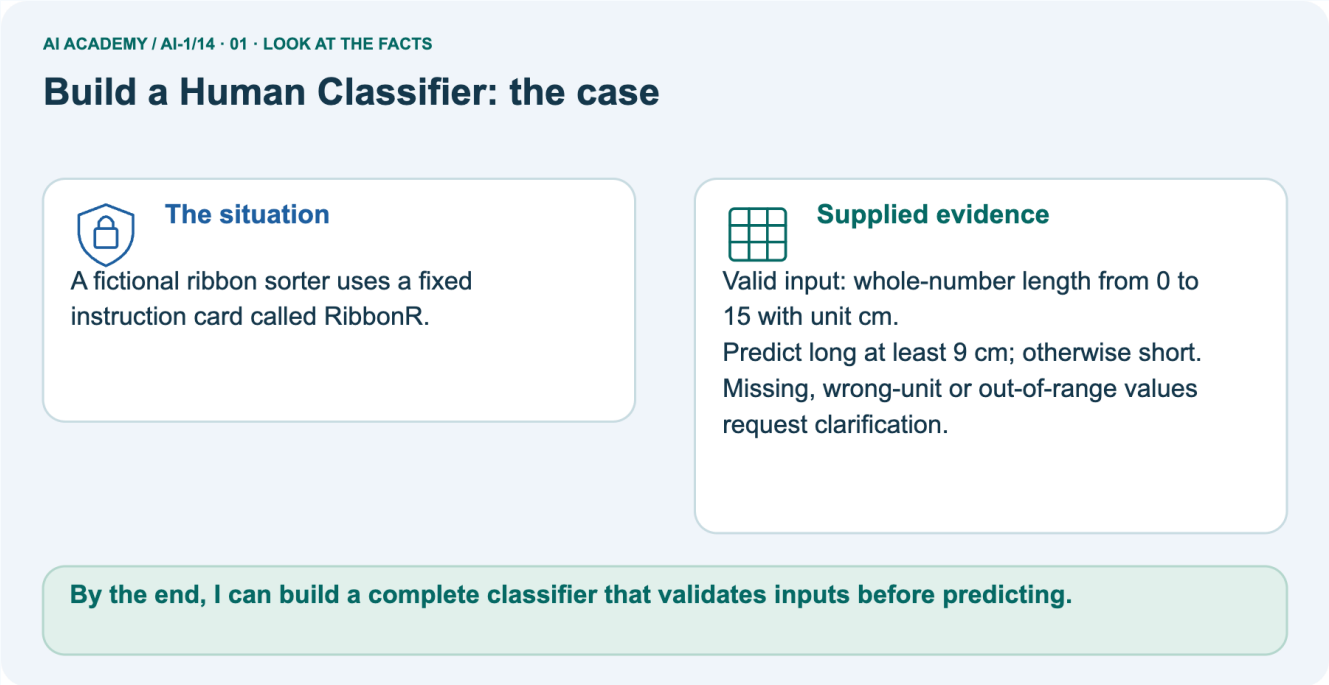
---

**STUDENT LESSON · 18 OF 18**

## Visual concept journey

> Explain each Build a Human Classifier graphic using its labels, deciding evidence and relevant limitation.

## 01. Build a Human Classifier: the case



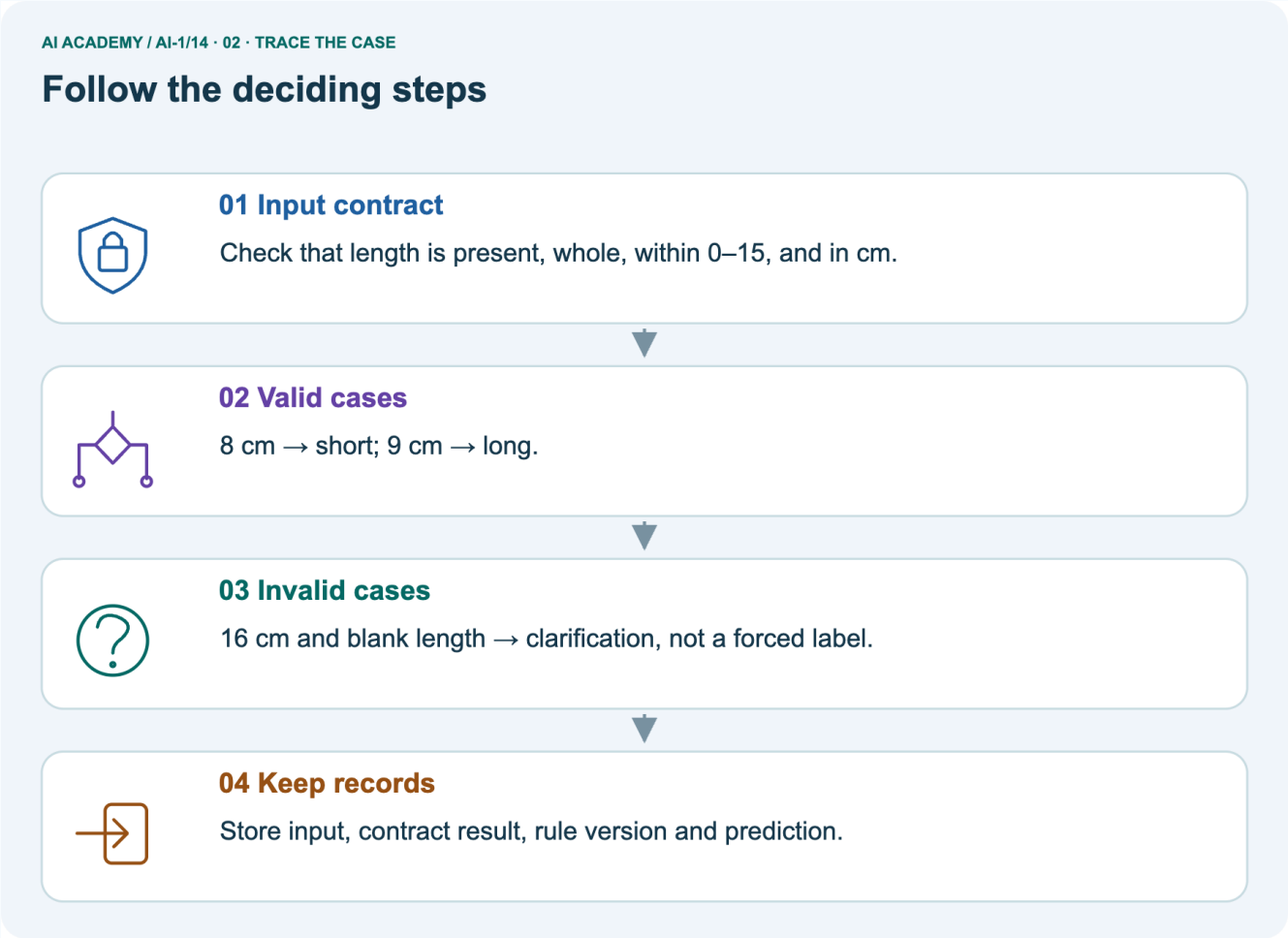

Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional ribbon sorter uses a fixed instruction card called RibbonR. Supplied facts: Valid **input**: whole-number length from 0 to 15 with unit cm. Predict long at least 9 cm; otherwise short.

Missing, wrong-unit or out-of-range values request clarification.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

**Input contract**: Check that length is present, whole, within 0–15, and in cm. → Valid cases: 8 cm → short; 9 cm → long. → Invalid cases: 16 cm and blank length → clarification, not a forced label. → Keep records: Store **input**, contract result, rule version and **prediction**.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. Three records through the Tower1 classifier

AI ACADEMY / AI-1/14 · 03 · INSPECT THE LESSON MODEL

## Three records through the Tower1 classifier

| Input record | Check and rule | Recorded outcome |
| --- | --- | --- |
| 5 cm | Accepted; 5 is below 6 | Tower1 predicts short |
| 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall |
| Height blank | Missing height; stop before rule | Tower1; no prediction |

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

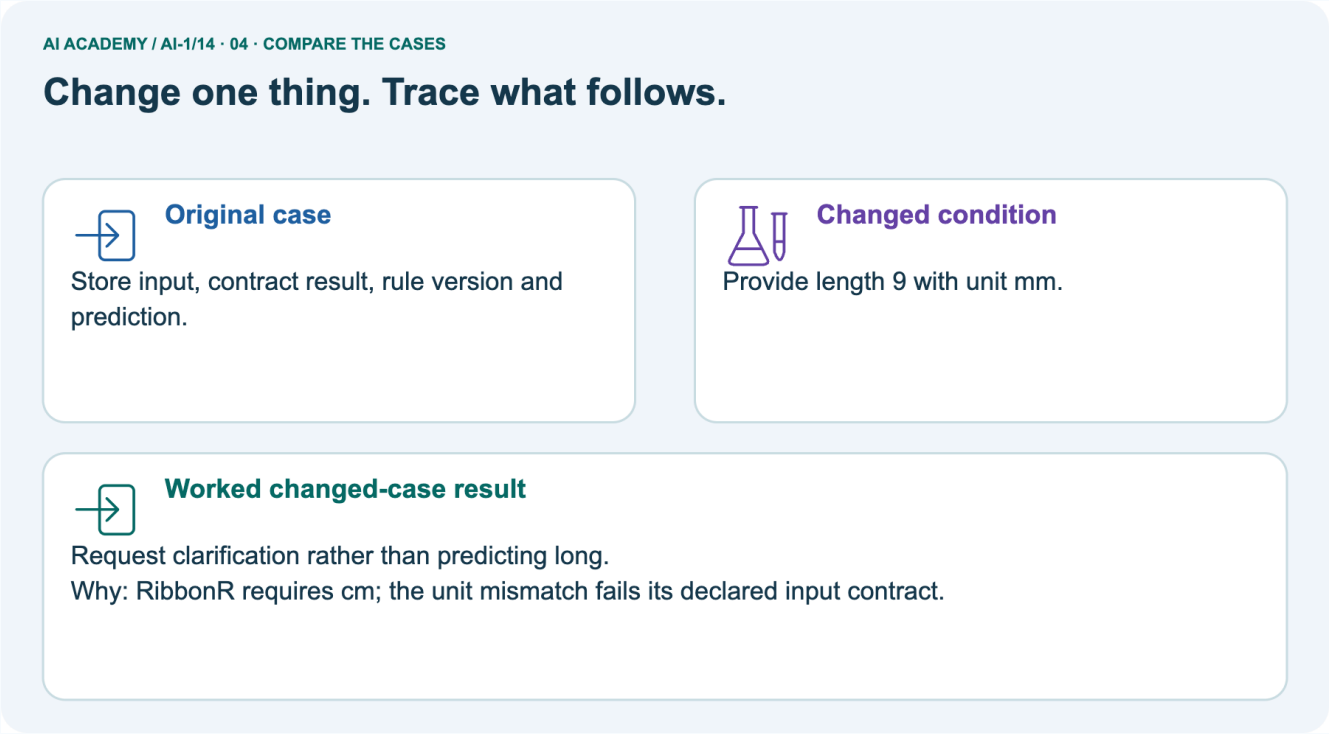
Follow the labelled picture with the matching explanation. Open the vector diagram

Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



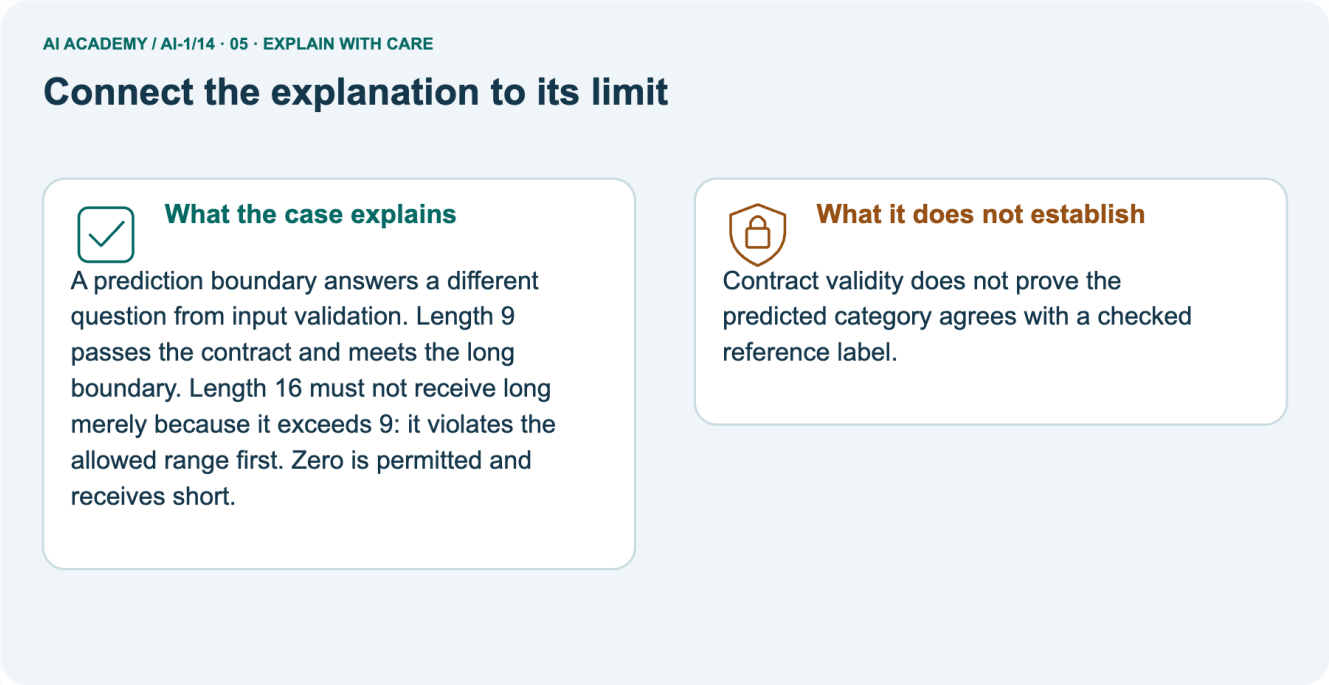

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Store **input**, contract result, rule version and **prediction**. Changed condition: Provide length 9 with unit mm. Expected result: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

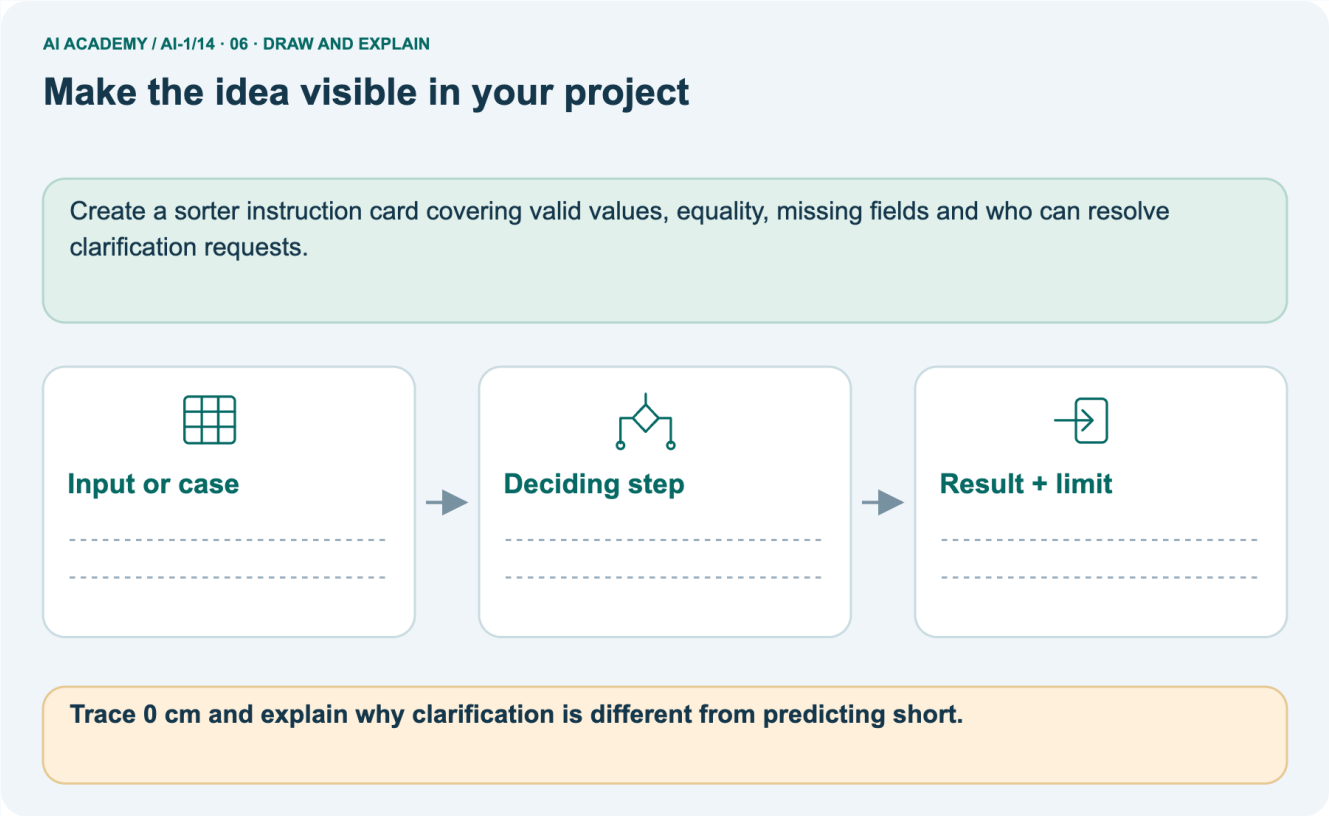
A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Limit: Contract validity does not prove the predicted category agrees with a checked reference label.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests. Draw your **input**, deciding step, result and limit.

Supported question: Trace 0 cm and explain why clarification is different from predicting short.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.